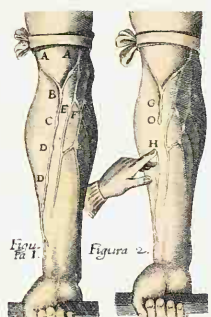
Blood circulation

AIR TO ARTERY
In this 13th-century illustration "life spirit" from inhaled air came into a dark lump in the center of the heart, where the arteries originated.

THE MUSCULAR HEART PUMPS BLOOD out through tubes, or vessels, called arteries. These branch repeatedly into microscopic capillaries that rejoin to form veins, which carry the blood back to the heart. This idea seems so obvious today that it is difficult to imagine any other way. Even before Aristotle (p. 6) and Galen (p. 9) produced their theories, people had all sorts of fanciful ideas to explain what the heart and vessels did. A Chinese account in the *Nei Ching* from almost 2,300 years ago, said: "The heart regulates all the blood… the current flows continuously in a circle and never stops." This was probably inspired guesswork, since the book had no anatomical evidence to prove it. After death, arteries collapse and seem to be empty, misleading many scientists. In ancient Greece, the arteries were thought to contain air and be part of the windpipe system. Coupled with the ancient beliefs in the four humors (p. 36), confusion reigned in explaining the functions of the heart, vessels, and blood. Some ideas had elements of truth. William Harvey's scheme, published in 1628, at last put science and medicine on the right track.

AROUND AND AROUND
The credit for establishing the circulatory system is usually given to English physician William Harvey (1578-1657), although Leonardo da Vinci came very close. After experiment and observation, in 1628 Harvey published *De Motu Cordis et Sanguinis in Animalibus*, or "On the Movement of the Heart and Blood in Animals." It stated that blood did not "ebb and flow" in the vessels as was widely accepted. It circulated around and around, pushed by the heart.



VEIN VALVES
Harvey based his ideas on careful study, rather than blindly following tradition. The approach of his work and his books were the beginning of the era of modern, scientific medicine. His illustrations show how blood in veins always flows toward the heart, as valves prevent it seeping in the wrong direction.

THE BLOOD VESSELS OF THE LEG
The branching nature of the circulatory system is clearly visible in the vessels in the leg. The external iliac artery brings high-oxygen blood into the limb. It divides into branches, which in turn subdivide and subdivide again, progressively becoming smaller. As microscopic capillaries, they deliver oxygen and nutrients to the tissues and collect waste products for removal. They rejoin, forming larger vessels to collect the blood, which join together and connect into the major veins. The external iliac vein is the main tube draining blood back to the heart.

SELF-BEATING HEART
Swiss-born Albrecht von Haller (1708-1777) was a poet, novelist, and botanist as well as anatomist, and he wrote an eight-volume study of physiology. He investigated how the arteries' muscular walls change the size of the vessels, so that the amount of blood flowing to different parts of the body could be varied.